

统计学习考试原题回忆

2022 级（2026 届）统计学 6 星

这张卷子的特点就是大的题目是中文的，里面的具体的细节是英文的，别奇怪中英混杂！

一、名词解释

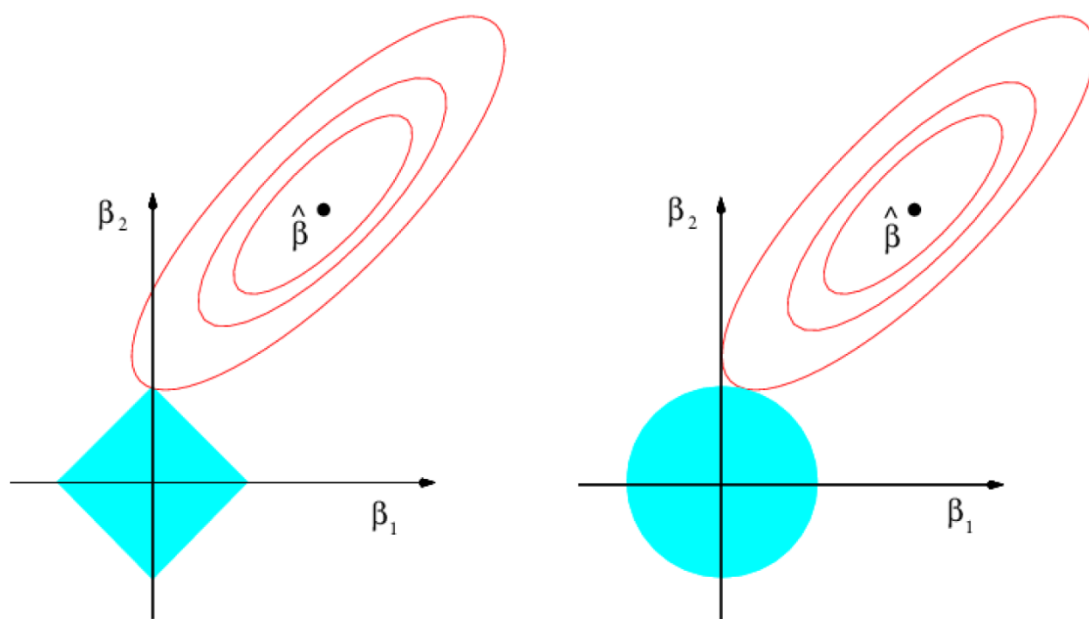
1. No Free Lunch Theorem
2. trade-off between model flexibility and interpretability
3. variable importance
4. Gini coefficient

二、简答题

1. Explain supervised learning and unsupervised learning, state their core objectives, and provide one typical application example for each.

2. Explain the basic idea of cross-validation, and compare K-fold cross-validation (K-fold CV) with Leave-one-out cross-validation (LOOCV) in terms of advantages, disadvantages, and practical applicability. （不确定！）

三、图表题



Interpret the figure above, compare the differences between the two methods, and explain the reasons for these differences.

四、计算题

Consider the problem of fitting a cubic spline to data. Derive its degrees of freedom when there are K interior knots. （不确定，但是是一道计算样条自由度的问题！）

五、证明题

1. Show that fitting a locally constant multinomial logit model of the form amounts to smoothing the binary response indicators for each class separately using a Nadaraya-Watson kernel smoother with kernel weights.

（这道题老师没讲，就发在群里面，我没看到，就没背.....）

2. In linear regression with Leave-one-out cross-validation (LOOCV), let y_i be the i -th observation, $\hat{y}_{(i)}$ be the prediction for y_i when the i -th sample is excluded, $\hat{y}_i$ be the fitted value from the full model, and s_i be the i -th diagonal element of the hat matrix (leverage). Prove that:

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{(i)})^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - s_i} \right)^2$$

（第二题有点拓展：1. 老师的证明里面是用的 $\mathbf{h}$ 作为对角元素，但是考试里面用的是 $\mathbf{s}$ ；2. 有两问，另一问是证明一个不等式，依稀记得是一个对角占优类似的条件，可以去 ESL 那本书上找原题）

找到了，是下面这道题（b、c 两小问合并了）！

Ex. 7.3 Let $\hat{\mathbf{f}} = \mathbf{S}\mathbf{y}$ be a linear smoothing of $\mathbf{y}$.

(a) If S_{ii} is the i th diagonal element of $\mathbf{S}$, show that for $\mathbf{S}$ arising from least squares projections and cubic smoothing splines, the cross-validated residual can be written as

$$y_i - \hat{f}^{-i}(x_i) = \frac{y_i - \hat{f}(x_i)}{1 - S_{ii}}. \quad (7.64)$$

(b) Use this result to show that $|y_i - \hat{f}^{-i}(x_i)| \geq |y_i - \hat{f}(x_i)|$.

(c) Find general conditions on any smoother $\mathbf{S}$ to make result (7.64) hold.

另外两道证明题是讲过的原题~